

BIOL 413 LAB (LM3)

PRINCIPLES OF BIOLOGY I

WHO*

Instructor: Dr. Kristen Johnson

Office: Room 569

Office Hours: Mondays 9-11am; Tuesdays 2-3pm; *and by appointment*

WHERE/WHEN

Lab (RM 540): THURS, 1-3:50pm

FEATURING · 4 LAB REPORTS/PRESENTATIONS (60 PTS) · 5 QUIZZES (25 PTS) · LAB EXAM (20 PTS) · ATTENDANCE/PARTICIPATION (20 PTS) ·

To do well in this lab:

- **Attend all labs! If you miss a lab, you will miss an experience that cannot be replicated by simply reading the lab.**
- **Be prepared when coming to lab. Most material is available in advance of the lab meeting time on the MyCourses/Canvas site.**
- **Make notes and sketches during lab. Ask questions, ask questions, ask questions!**
- **Fill out lab reports thoroughly, and answer questions with more than a word or two.**
- **Study for the practical exam *every week* by reviewing all handouts, lab reports, notes & sketches. If you wait until just before the practical, it will be too much information.**

REQUIRED MATERIALS

LAB COURSE PAGE

All of the materials for Bio 413 lab are found on Canvas. You will use Canvas materials to review any videos or interactive programs that we will use in lab as well as print out some materials needed for lab.

OTHER STUFF

Lab Notebook
Lab Coat

*CONTACT INFO

kristen.johnson@unh.edu
603-641-4177 (office)

BIOL 413 LM3: LAB COURSE SCHEDULE (JOHNSON)

Week	Date	Lab Activity	Quiz	Report/Assignment Due
Module 1: Scientific Method				
1	31-Aug	Lab Introduction & Safety, Introduction to Brine Shrimp and Scientific Method		
2	7-Sept	Experimental Design		
3	14-Sept	Run Experiment, Data Collection, Statistical Analysis		
4	21-Sept	Communicating in Science; Anatomy of a Journal Article, Conventions, Presentations	Quiz #1: Brine Shrimp	
5	28-Sept	Student Presentations		Module 1: Student Presentations
Module 2: Cells				
6	5-Oct	Mitosis, Meiosis and Cell Division		Module 2: Lab Report
Module 3: Molecular Genetics				
7	12-Oct	Crime Scene: Evidence Collecting & Blood Typing	Quiz #2: Mitosis/Meiosis	
8	19-Oct	Crime Scene: ELISA Test	Quiz #3: blood typing	
9	26-Oct	Crime Scene: Restriction Fragment Analysis	Quiz #4: ELISA	
10	2-Nov	Crime Scene: Final Analysis & Communicating in Science – Forensic report	Quiz #5: RFLP	
11	9-Nov	Veteran's Day (No Labs)		
12	16-Nov	Student Presentations		Module 3: Student Presentations
13	23-Nov	Thanksgiving (No Labs)		
Module 4: Metabolism				
14	30-Nov	Metabolism		Module 4: Lab Report
15	7-Dec	LAB EXAM		

Detailed Methods of Testing/Evaluation: The course grade will be based on the student's performance on the following:

Method of Evaluation	Point Value
Presentation/Lab Report from Module 1 & 3 (20 pts ea)	40
Lab Report from Module 2 & 4 (10 pts ea)	20
Quizzes (5 pts ea)	25
Lab Exam	20
Attendance/Participation	20
Total Points Possible	125

THE DO's and DON'Ts...The Fine Print

Attendance/Participation:

Attendance in lab is very important as each lab of a module builds upon another. If you miss a lab, you will be responsible for gathering the information needed to come to the next lab ready to perform the lab of that week. Each week, we will be working as groups, and it is not fair to the others in your group to miss lab as that will leave the other members of the group more work to do. In light of these facts, **5 pts will be deducted from your lab grade for each missed lab**. Any more than 2 missed labs will result in possible failure of the lab. Also, points may be deducted if you are not a contributing member of your team for the presentations.

Lab Classroom Conduct:

- *Cell phones:* Please **turn off/silence** and put away your cell phone before you enter the laboratory. If you must use your phone, please step out of the lab.
- *Computer Use:* You may have an electronic device (laptop/tablet) out *during lab* for lab-related activities **ONLY** when instructed by your professor. You should **NOT** be using electronics for social networking, emailing, playing games, etc. Such behavior distracts you and your classmates from the lab exercises and is considered disrespectful to the professor and your classmates. In addition, chemicals from our experiments may contaminate your electronic device.
- *Food:* There is a strict policy of **NO** food or drinks in the laboratory.

MAKE-UP QUIZZES/EXAMS:

- It is expected you will be present to take all lab quizzes. Make-up lab quizzes will be given **ONLY** if it is missed for a valid (documented) reason and discussed ahead of time.

ACADEMIC HONESTY

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<http://unh.edu/vpsas/handbook/academic-honesty>). By turning in any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual homework assignments, or allowing your own work to be used by another student; copying information from a web site or other text without proper documentation; buying a pre-written paper or lab report. *Your assignments submitted through Canvas will be routinely screened for plagiarism using sophisticated software.*

Cheating is mainly concerned with copying on quizzes, exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and placed upside down on the table in front of you for the duration of the exam. Ear buds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

ACADEMIC SUPPORT/TUTORING

The Center for Academic Enrichment (CAE) [in the 2nd floor Learning Commons] is a resource for all students enrolled in UNH Manchester courses. Their mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. Support for other content areas is dependent on the background and competencies of current peer tutor staff. All enrolled students are entitled to 1 hour of free one-to-one tutoring per week, per course. In addition, peer and professional tutors may run study groups for your class and provide drop-in math tutoring. Please visit the CAE online at <http://manchester.unh.edu/student/center-academic-enrichment> to learn more about their services or to make an appointment with a tutor. **Writing Tutors:** It is also strongly recommended that you take advantage of the CAE writing tutors to turn in high quality scientific writing in the lab. Sometimes students do well on the exams and quizzes and poor paper writing brings down their grade—your goal is to learn scientific writing and style to prepare you for the second semester and beyond, where you will be expected to have already mastered the (basic) writing concepts you learn in this course.

DISABILITIES SERVICES

The University is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must register with the Disability Services Office. The Disability Services Coordinator at UNHM is Jenessa Zurek. Jenessa can be contacted at (603) 641-4170, jenessa.zurek@unh.edu, or in person in the Student Services Suite, Room 410H.